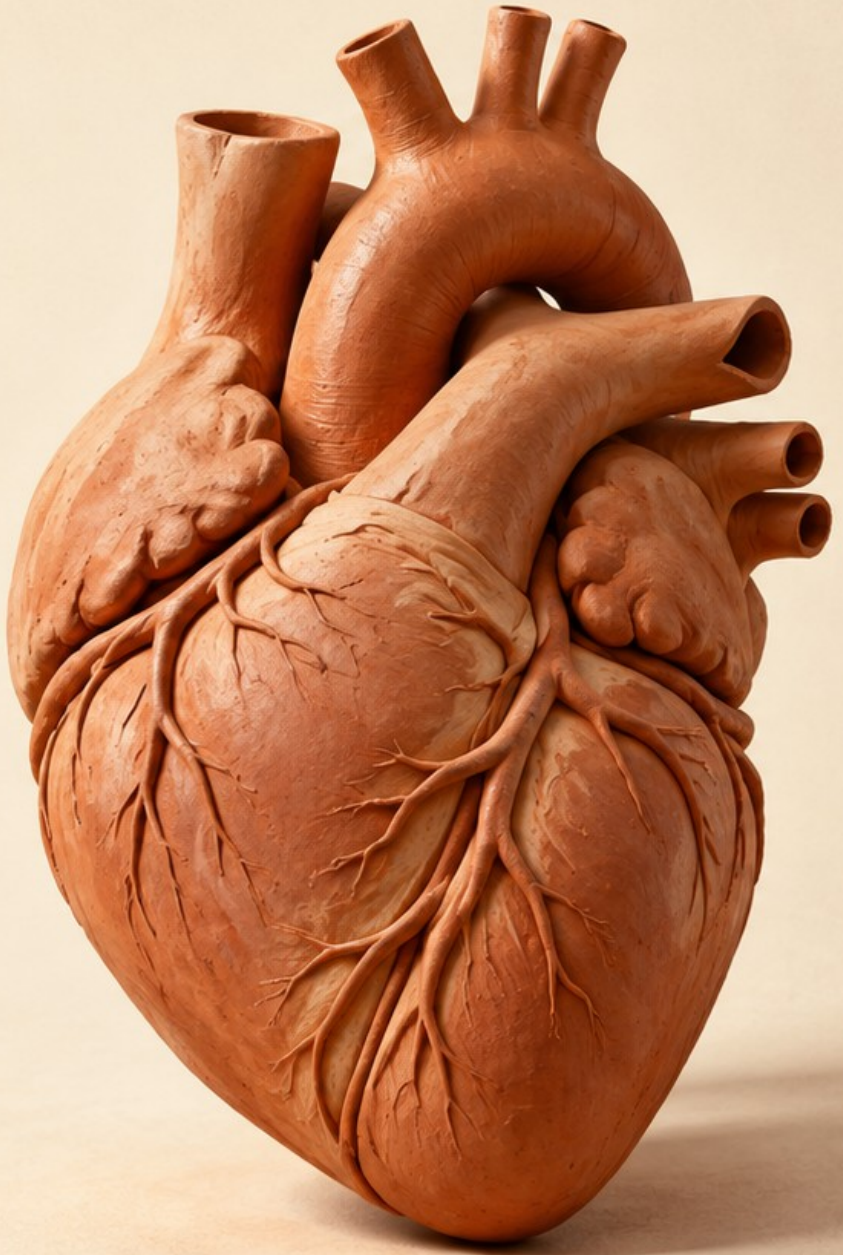
Heart-disease classification

Conventional ML and a provisional Bayesian Network

Phase 1 project presentation
Team T008, Section C

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Project scope

The task

Classify recorded disease presence from eleven clinical predictors.

The comparison

Four conventional classifiers and a manually specified Bayesian Network.

The research question

Which model performs best under a shared development F1 rule, and how does the chosen procedure transfer across hospitals?

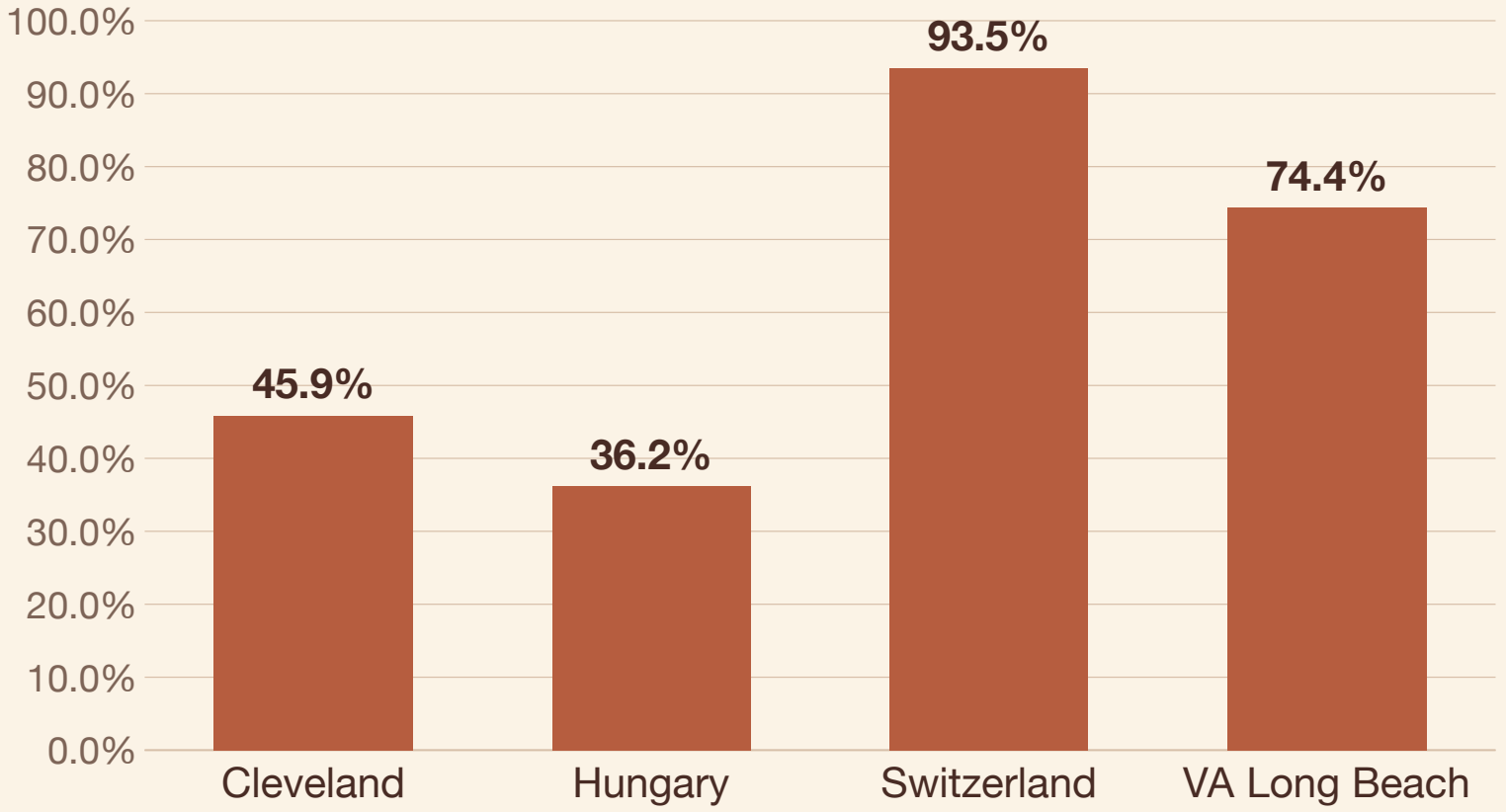
Task: historical disease presence, defined as num > 0

A balanced total hides hospital differences

918

retained records
from 920 supplied rows

508 positive
410 negative



Exact-record deduplication removes two records. Patient linkage remains unknown.

Source: verified project run, 25 October 2026

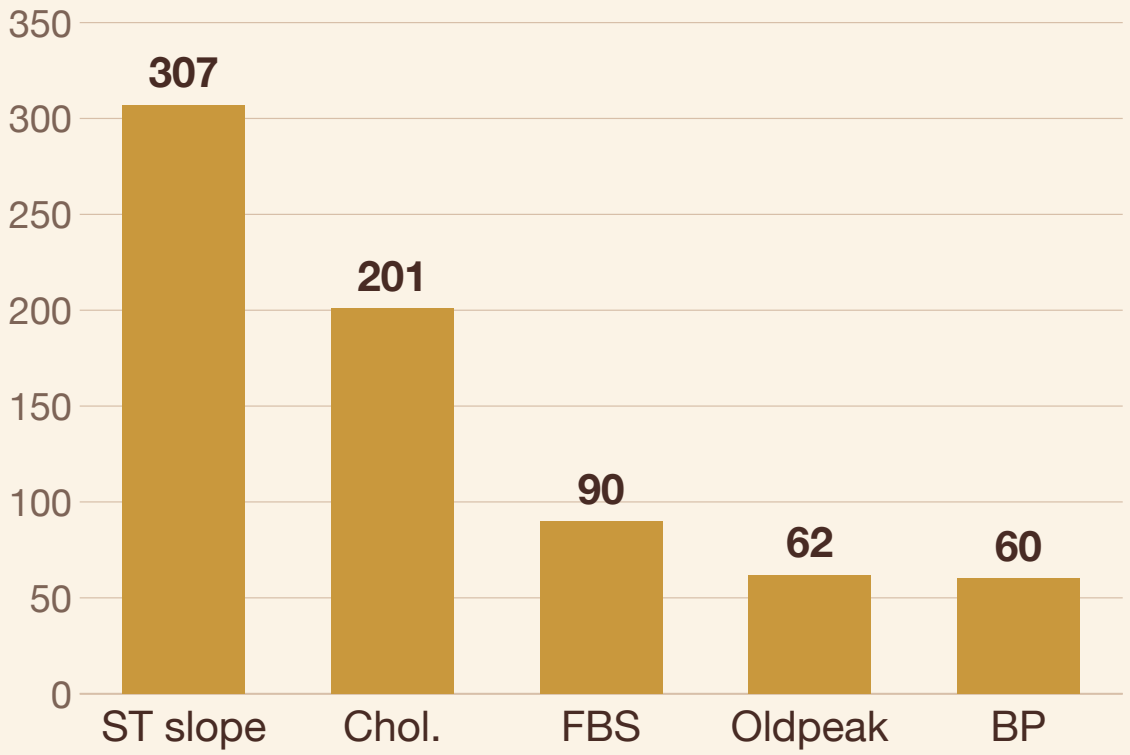
Missingness shapes the feature pipeline

Eleven eligible predictors

5 numeric and 6 categorical fields.
Exclude ID, hospital and target.
Exclude ca and thal for severe missingness.

Learned inside every training fold

Numeric median imputation and indicators, then scaling. Categorical mode imputation and one-hot encoding.



Missing records after cleaning
FBS: fasting sugar. BP: blood pressure.

Source: verified project run, 25 October 2026

Separate selection, cutoffs and evaluation

Partition	Records	Permitted role
Selection S	550	Nested tuning and model fitting
Cutoff C	184	Threshold selection on frozen models
Retrospective T	184	Evaluation after choices freeze

5 outer folds

3 inner folds

Common rows and folds across families. Training rows alone fit all learned transforms.

Source: verified project run, 25 October 2026

Five models provide different assumptions

Model	Learning mechanism	Search configurations
Logistic Regression	Regularized linear log-odds	8
Random Forest	Average randomized decision trees	6
Histogram Gradient Boosting	Sequential trees on binned features	8
XGBoost	Regularized sequential tree boosting	8
Bayesian Network	Factorized probabilities and CPTs	1

Different search budgets limit claims about universal model superiority.

Source: verified project run, 25 October 2026

The Bayesian Network remains provisional

Structure

Disease conditions all eleven features.
Oldpeak additionally conditions ST slope.

Training

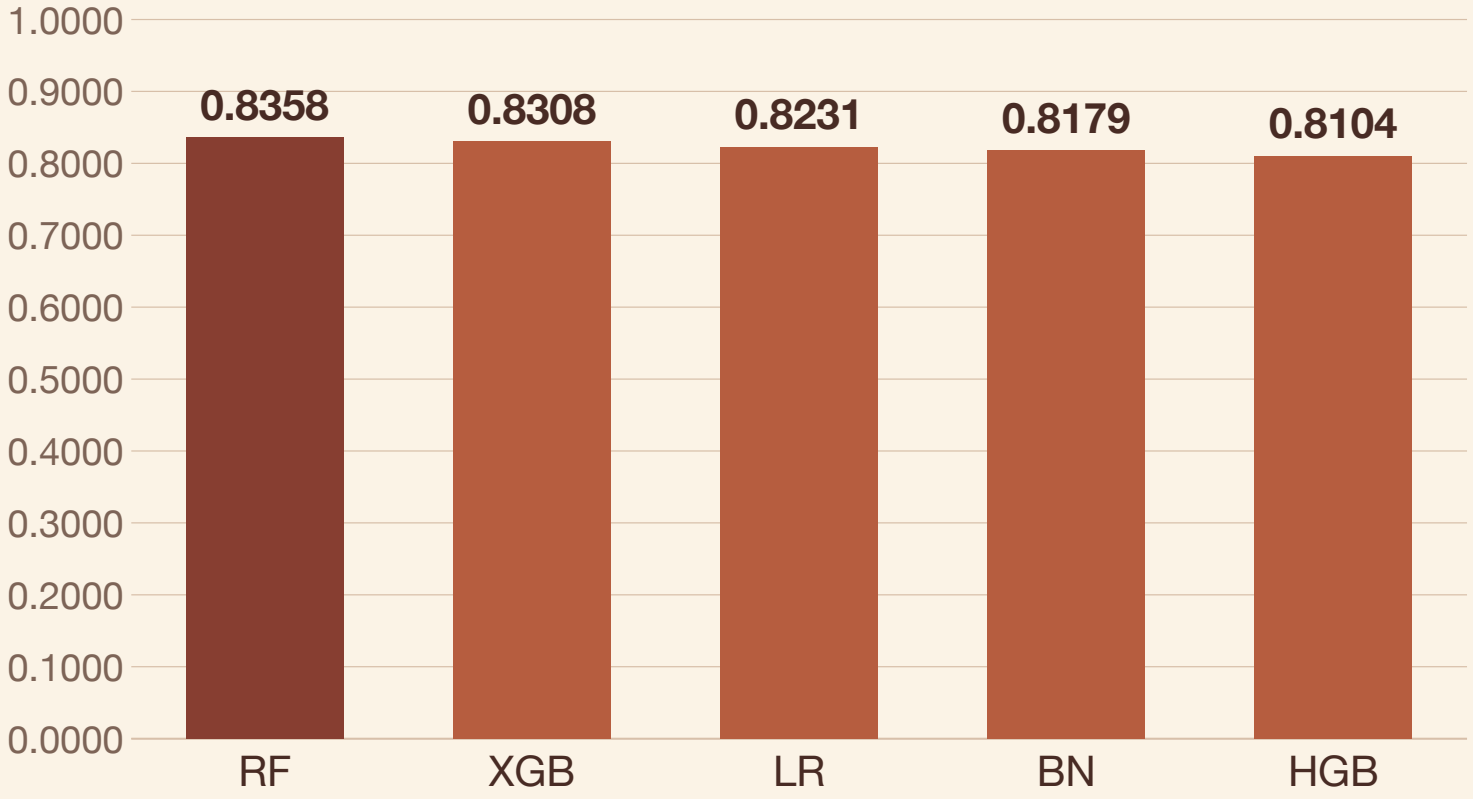
Learn quantile bins and conditional probability tables on training rows. Alpha = 1 smooths state counts.

Inference and limitation

Sum over missing query evidence. The graph assumes many conditional independences that need domain justification.

Source: verified project run, 25 October 2026

Random Forest leads development F1



0.8358

Highest mean validation F1 under the predefined rule

Only 0.0049 above XGBoost.
No formal superiority claim.

Family selection happens before cutoff or retrospective evaluation scoring.

Source: verified project run, 25 October 2026

Logistic Regression leads several later metrics

Model	Accuracy	Precision	Recall	F1	AUC
Logistic Regression	**0.8315**	**0.8257**	**0.8824**	**0.8531**	**0.9125**
Random Forest	0.8043	0.7538	0.9608	0.8448	0.9029
HGB	0.8152	0.7698	0.9510	0.8509	0.8945
XGBoost	0.8043	0.7463	0.9804	0.8475	0.9039
Bayesian Network	0.7880	0.8058	0.8137	0.8098	0.8765

LR: F1 0.8531 and AUC 0.9125. RF remains the development-selected family.

Retrospective T: 184 records, including 102 positive labels

Random Forest detects 98 of 102 positives

Actual / predicted	Negative	Positive
Negative	50 true negatives	32 false positives
Positive	4 missed positives	98 true positives

96.08%

recall at cutoff 0.38

84.48% F1
90.29% ROC-AUC

The cost of high recall: 32 of 82 negative records receive positive decisions.

Source: verified project run, 25 October 2026

Why Random Forest was a strong candidate

200 randomized trees

Different samples and feature subsets diversify trees. Averaging can reduce variance.

Depth limited to 5

Shallow trees and a minimum leaf size of 2 constrain each tree’s complexity.

Nonlinear relationships

Trees can represent interactions without assuming linear log-odds. This made the forest a useful candidate for tabular data.

These mechanisms plausibly explain competitiveness. No ablation isolates their effect.

Source: verified project run, 25 October 2026

A lower cutoff trades specificity for recall

RF operating point	Recall	Specificity	Missed positives	False positives
Default cutoff 0.50	0.8725	0.7195	13	23
C-selected cutoff 0.38	0.9608	0.6098	4	32

The same model changes its decisions when the operating cutoff changes.

F1 optimization does not encode clinical costs. Both error types must remain visible.

Source: verified project run, 25 October 2026

Model strengths come with different costs

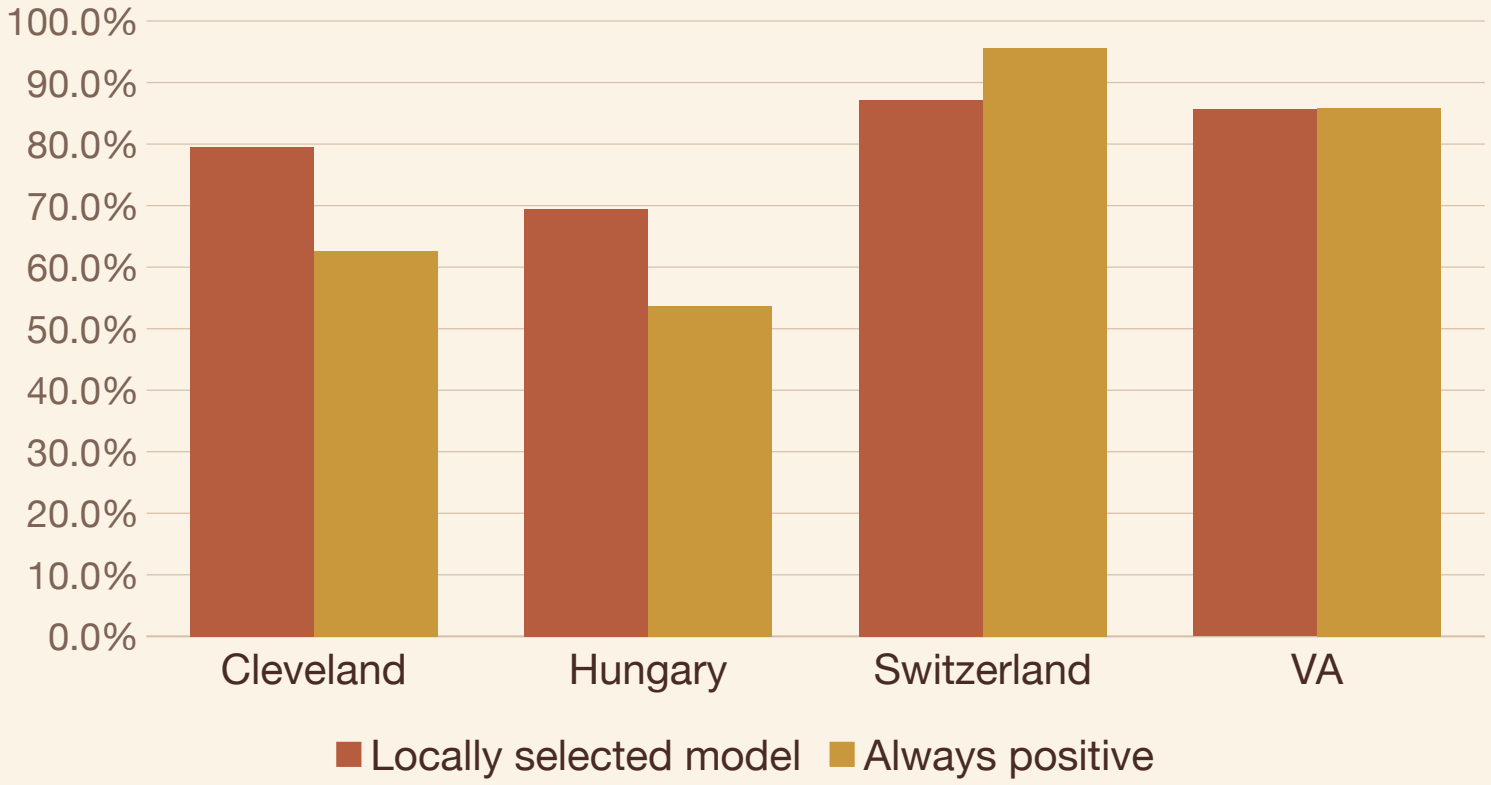
Model	Why it is useful	Main trade-off
Logistic Regression	Compact explanation and strong later scores	Linear log-odds can miss interactions
Random Forest	Nonlinear patterns and averaging of trees	Larger model, less direct interpretation
HGB / XGBoost	Sequential correction with regularization	More tuning choices and dependencies
Bayesian Network	Explicit probabilities and missing-evidence queries	Binning and restrictive graph assumptions

Performance depends on the objective, cohort, threshold and validation design.

Source: verified project run, 25 October 2026

Hospital transfer reveals important failures

F1 on each excluded hospital's development records



5%

VA specificity
2 of 40 negatives rejected

Switzerland: only 8 negatives.
Its selected RF trails the always-positive F1 reference.

A high positive-class F1 can hide weak negative-class decisions.

Source: verified project run, 25 October 2026

Challenges and the responses we implemented

Challenge	Implemented response	Remaining limitation
Missing and site-dependent data	Cleaning and fold-local imputation	Measurement bias may remain
Leakage during tuning	Nested CV and separate cutoff records	Family-selection optimism remains
High recall with false alarms	Report confusion matrices and specificity	Clinical error costs are unknown
Domain graph and prior test exposure	Explicit assumptions and retrospective labels	BN domain adequacy remains provisional

Source: verified project run, 25 October 2026

The results support a qualified conclusion

Selected by the protocol

Random Forest wins development F1.
Its 0.38 cutoff finds 98 of 102 positives.

Strong comparator

Logistic Regression leads several retrospective metrics. The chosen objective determines what “best” means.

Evidence and boundary

48 tests pass. 25 saved models verify. Hospital shift and a provisional BN graph limit generalization claims.

Phase 2 adds the assigned neural-network comparison.

Source: verified project run, 25 October 2026

References and project evidence

Dataset	Janosi et al. UCI Heart Disease (1989). DOI: 10.24432/C52P4X
Random Forest	Breiman (2001). DOI: 10.1023/A:1010933404324
Bayesian classifiers	Friedman, Geiger and Goldszmidt (1997). DOI: 10.1023/A:1007465528199
XGBoost	Chen and Guestrin (2016). DOI: 10.1145/2939672.2939785
Calibration	Van Calster et al. (2019). DOI: 10.1186/s12916-019-1466-7
Validation and HGB	scikit-learn documentation: nested CV and ensemble methods

Shared repository: nishtha-09-gupta / AML_Capstone_Project
Commit 16f1e33, verified experiment dated 25 October 2026

Full citations and experiment details appear in the IEEE report